

Heat, cold and first hospitalisation after cardiovascular diagnosis in Hong Kong

An exploratory monthly study among people with diabetes or hypertension, 2013–2023

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Summary

Hong Kong is becoming hotter, especially at night, but cold winter days have not disappeared. Both heat and cold can place additional stress on the cardiovascular system. This matters particularly for people living with type 2 diabetes or hypertension, who already face a high risk of coronary heart disease and heart failure.

This project examined whether monthly temperature patterns were associated with monthly counts of a carefully defined hospital event. The available Hospital Authority data counted each person's first recorded hospitalisation after their first diagnosis record for coronary heart disease (CHD) or heart failure (HF). They did not record the reason for admission. The study therefore does not claim that every admission was caused by CHD or HF, and it does not estimate individual risk. Its question is narrower: in months with more heat or cold, did the governed territory-wide count of these first hospitalisations also differ?

The analysis covered January 2013 to December 2023, giving 132 monthly observations. Six temperature measures were examined separately for each outcome: monthly mean temperature, mean daily maximum temperature, mean daily minimum temperature, hot nights, cold days, and very hot days. This produced a panel of twelve comparisons. Models accounted for season, long-term change, and the different number of days in each month. Four approaches to uncertainty were shown, and the twelve comparisons were corrected for multiple testing.

Two associations stood out before that correction. Five additional hot nights in a month corresponded to a CHD count ratio of 1.022 (95% confidence interval 1.002–1.042 using Newey–West lag 6). Five additional cold days corresponded to an HF count ratio of 1.073 (1.006–1.144). These are approximately 2.2% and 7.3% higher monthly counts, respectively. However, both multiple-testing-adjusted q -values were 0.192, and all twelve q -values exceeded 0.19. The CHD hot-night interval also included no association under two of the four uncertainty methods. The findings are therefore exploratory, not confirmatory.

The main contribution is not a claim that one weather measure has been proved to cause one disease. It is a reproducible framework that reports the complete panel, makes uncertainty visible, and refuses claims that the available data cannot support. Better daily outcome data, a denominator for the cohort still at risk, and a recorded cause of admission would be needed for stronger inference.

1. Introduction

1.1 A city facing heat and cold

Climate change does not replace one thermal hazard with another. In Hong Kong, hot nights have become more frequent, while cold days still return each winter. At Hong Kong Observatory Headquarters, the number of hot nights rose from 10 in 2013 to a peak of 61 in 2021. Cold-day counts fluctuated substantially but remained present throughout the study period. The city must therefore plan for a changing mixture of overnight heat, very hot days, and episodic cold rather than assuming that warming makes cold irrelevant.

Temperature can affect cardiovascular health through several pathways. Heat can promote dehydration, disturb sleep and overnight recovery, and increase cardiac demand. Cold can raise blood pressure and vascular resistance. Responses may be stronger in people whose cardiovascular regulation is already compromised by diabetes or hypertension. These mechanisms make both sides of the temperature distribution relevant, but they do not establish what will be visible in a particular dataset.

Hong Kong already has important evidence on daily temperature and health. Earlier studies examined daily stroke or acute myocardial infarction admissions and reported substantial cold-related patterns (Goggins et al. 2012; Goggins, Chan, and Yang 2013). Liu and colleagues estimated mortality attributable to non-optimal temperature and found a larger fraction associated with cold than heat during 2006–2016 (J. Liu et al. 2020). More recent work modelled heat-related excess deaths under several heatwave definitions (Z. Liu et al. 2026). Those studies answer valuable but different questions. A daily mortality attributable fraction, a modelled number of excess deaths, and a monthly hospitalisation count ratio cannot be substituted for one another.

1.2 Why monthly data require a different question

Health data are often released in aggregate form to protect privacy. In this project, the governed outcomes were available only as territory-wide monthly counts. That resolution can support a study of monthly burden, but it cannot reveal whether a particular hot day triggered a hospitalisation several days later. It also cannot show which individual experienced which temperature.

The outcome definition requires equal care. The delivered data count the first recorded hospitalisation after a first CHD or HF diagnosis record among people with type 2 diabetes and/or hypertension. Admission cause is absent. Calling these events “heart attacks caused by temperature,” or even “principal-diagnosis CHD admissions,” would go beyond the data. The report uses the longer but accurate description because precision about the outcome is part of the science.

1.3 Objectives

The primary objective was to estimate monthly associations between thermal burden and first-hospitalisation counts for CHD and HF during 2013–2023. Three secondary objectives were to:

1. compare continuous temperature measures with official extreme-day counts;
2. show how conclusions change across defensible uncertainty methods; and
3. determine whether the available monthly outcomes could reliably support a more ambitious estimate using daily exposure information.

2. Methods

2.1 Study design and data

The study used an ecological monthly time-series design. Each observation was one calendar month for Hong Kong, from January 2013 through December 2023. The final panel contained 132 months.

Daily weather observations came from Hong Kong Observatory Headquarters. Public weather series were checked against annual Observatory summaries before analysis. Population data came from the Census and Statistics Department, and air-pollution series were assembled from Environmental Protection Department general monitoring stations for the broader project. The core models reported here focus on thermal exposures and do not claim that pollution or influenza confounding has been fully resolved.

Governed Hospital Authority aggregates supplied the two outcomes. The cohort comprised people recorded with type 2 diabetes and/or hypertension during the study period. For each outcome, the monthly series counted the first recorded hospitalisation after the person's first CHD or HF diagnosis record. The analysis used aggregate counts only; source monthly health files and merged health panels were not placed in public version control.

2.2 Temperature measures

No single number fully represents thermal burden. A warm monthly average can arise from consistently mild conditions, several intense days, or unusually warm nights. These patterns may matter differently for health. Six measures were therefore evaluated:

- monthly mean temperature;
- monthly mean of daily maximum temperature;
- monthly mean of daily minimum temperature;
- number of hot nights, defined as minimum temperature at least 28 °C;
- number of very hot days, defined as maximum temperature at least 33 °C; and
- number of cold days, defined as minimum temperature at most 12 °C.

The continuous measures were interpreted per 1 °C. Extreme-day counts were interpreted per five additional days in a month. More complex spell and heatwave definitions were developed and source-checked elsewhere in the project (Wang et al. 2019; Li et al. 2025), but the six-measure panel provides the clearest common comparison for this report.

Cold-day identification used between-year variation within the same calendar month and was concentrated in December–February. A winter-only restriction is a future sensitivity analysis, not a cure for identification.

Official thermal extremes at HKO Headquarters, 2013–2023

Annual counts. Pipeline totals match HKO Year's Weather summaries (33/33).



Source: Hong Kong Observatory dailyExtract, Headquarters station. Environmental descriptors only — not health-outcome results.

Figure 1: Official hot nights, very hot days and cold days at Hong Kong Observatory Headquarters, 2013–2023. These are environmental descriptors, not health findings.

2.3 Statistical analysis

Separate negative-binomial regression models were fitted for each outcome and each exposure: two outcomes multiplied by six exposures produced twelve core comparisons. Negative-binomial models allow monthly counts to vary more than a simple Poisson model would permit. Each model controlled for calendar month, which captures recurring seasonal patterns, and a natural spline of time with four degrees of freedom ($ns(\text{time}, 4)$). The number of days in the month was included as an offset so that February was not treated as providing the same observation time as a 31-day month.

The reported effect measure is a **count ratio**. A ratio of 1 means that the model estimates no difference in the monthly count for the stated exposure change. A ratio of 1.05 means an estimated 5% higher count; a ratio of 0.95 means an estimated 5% lower count. Because the number of people in the cohort who remained eligible for a first event was unavailable for each month, these are count ratios rather than individual incidence-rate ratios.

Monthly observations close together in time may remain correlated after season and trend are modelled. To avoid hiding this choice, uncertainty was reported four ways: model-based, heteroskedasticity-consistent (HC1), and Newey–West methods with lags of three and six months. The continuity analysis used the six-month Newey–West interval, but all four were displayed. No method was chosen because it made a confidence interval exclude 1.

In the leading models, Pearson residual autocorrelation at lag 1 was 0.508 for CHD hot nights and 0.146 for HF cold days.

Twelve comparisons also create more opportunities for a small p-value to occur by chance. Benjamini-Hochberg q-values were therefore calculated across the complete twelve-comparison panel. A q-value is not the probability that a result is true or false. It is a tool for limiting the expected proportion of false discoveries when a set of comparisons is considered together.

2.4 Robustness and the attempted daily-exposure extension

Additional analyses varied the long-term trend, removed early years, restricted the series to the pre-2020 period, examined exposure lags, and excluded the most influential month. These checks ask whether a result depends on a narrow modelling choice or one observation. Joint models were used only to diagnose what happens when correlated heat measures compete in the same equation.

The project also evaluated a constrained method that combines monthly outcomes with daily exposures (Basagaña and Ballester 2024). Before it could be used on real health outcomes, it had to recover known effects in simulations designed around Hong Kong weather and the observed monthly setting. The F1.1 500-replicate run failed its admission criteria. After that run, F1.2 re-summarised the existing fits against unchanged numerical thresholds using worst-cell rules; admission remained fail. No daily health coefficient was therefore produced. This project-specific result does not show that the Basagaña-Ballester method fails generally.

3. Results

3.1 Environmental and outcome patterns

The weather record shows a clear rise in hot nights and very hot days across the decade, with year-to-year variation. Cold days were less common than hot nights in recent years but remained recurrent. This coexistence supports examining heat and cold together rather than treating one as a historical concern.

The governed outcome series contained 156,156 CHD first-hospitalisation events and 29,681 HF events across 132 months. The corresponding monthly means were 1,183.0 and 224.9. Both outcomes displayed seasonality, with a stronger winter peak for HF. These totals describe the study series; they do not establish an effect of weather.

Table 1. Outcome summary

Outcome	Months	Total events	Mean per month	Event definition
Coronary heart disease	132	156,156	1,183.0	First hospitalisation after first recorded CHD diagnosis
Heart failure	132	29,681	224.9	First hospitalisation after first recorded HF diagnosis

3.2 The twelve-comparison panel

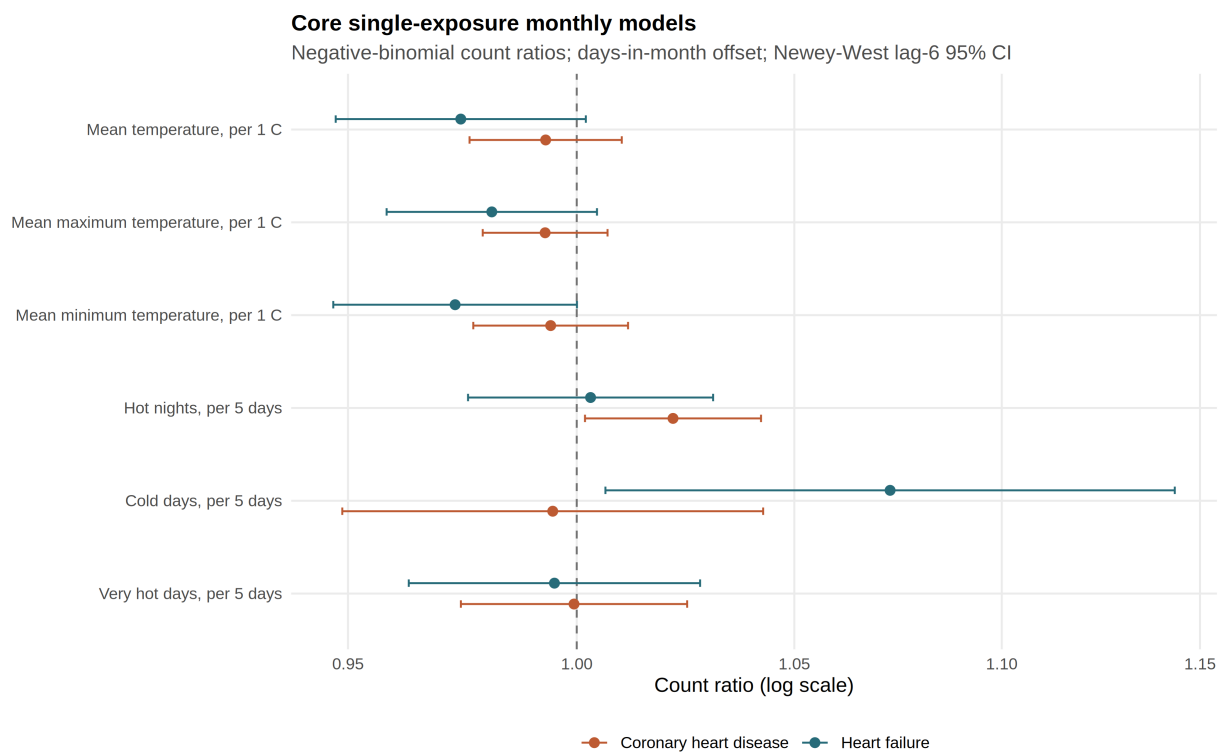


Figure 2: Count ratios and 95% confidence intervals for the twelve core monthly models. The dashed line at 1 indicates no estimated difference. All twelve Benjamini-Hochberg q-values exceeded 0.19.

Most core estimates were close to a count ratio of 1. For CHD, monthly mean, maximum and minimum temperatures produced count ratios of approximately 0.993–0.994 per 1 °C. Cold days and very hot days were also near 1. The largest CHD estimate was for hot nights: 1.022 per five additional hot nights (95% confidence interval 1.002–1.042 using Newey-West lag 6; $p = 0.032$).

For HF, the continuous temperature estimates were modestly below 1. The minimum-temperature estimate was 0.973 per 1 °C (0.947–1.000). Hot nights and

very hot days were near 1. The largest HF estimate was for cold days: 1.073 per five additional cold days (1.006–1.144; $p = 0.031$).

In plain language, the two leading estimates correspond to approximately 2.2% more CHD first-hospitalisation events per five additional hot nights and 7.3% more HF first-hospitalisation events per five additional cold days. These descriptions concern modelled monthly counts, not an individual’s probability of hospitalisation.

After correction across all twelve comparisons, the two leading estimates each had $q = 0.192$. Every core q -value exceeded 0.19. The complete panel therefore provides no multiplicity-protected confirmatory association.

3.3 What the uncertainty comparison changed

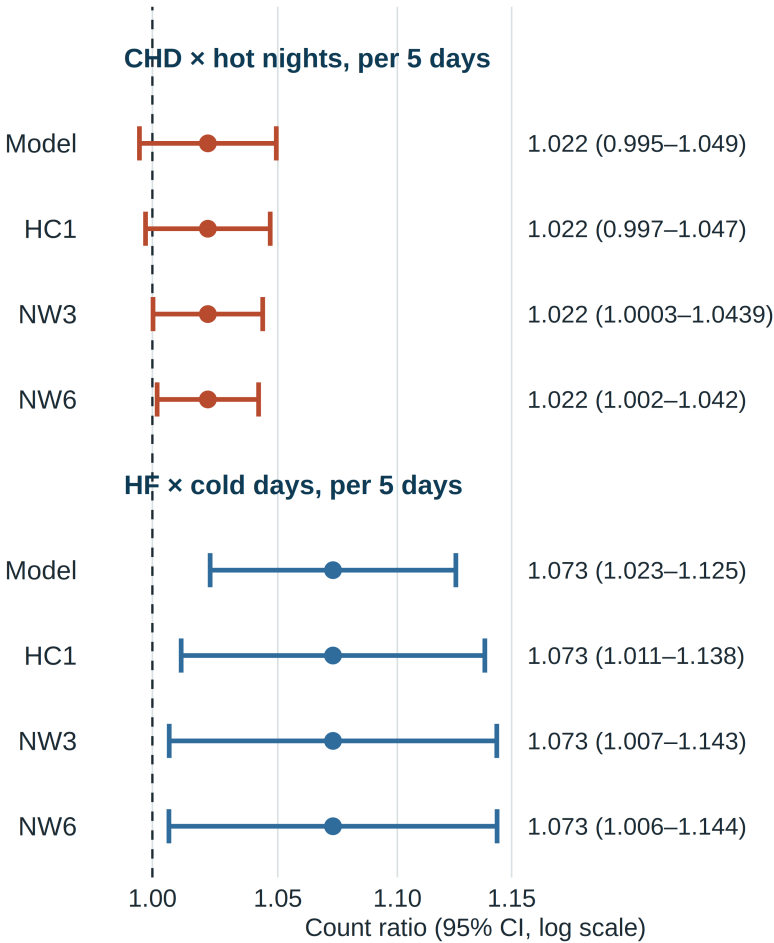


Figure 3: The two leading exploratory estimates under four approaches to standard errors. The CHD hot-night interval crosses 1 under two methods; the HF cold-day interval does not.

The CHD hot-night point estimate was 1.022 under all four standard-error methods. The intervals were 0.995–1.049 using model-based uncertainty, 0.997–1.047 using HC1, 1.0003–1.0439 using Newey-West lag 3, and 1.002–1.042 using Newey-West lag 6. Model-based and HC1 intervals included 1; NW3 excluded 1 only on the unrounded scale, and NW6 excluded 1. No method was selected because its interval excluded the null.

The HF cold-day point estimate was 1.073. Its four intervals were 1.023–1.125, 1.011–1.138, 1.007–1.143, and 1.006–1.144. All excluded 1, so this association was more stable across uncertainty methods. That stability does not overcome the multiple-testing result: its q-value remained 0.192.

3.4 Robustness and calibration

Across alternative trend and early-period specifications, the CHD hot-night count ratio ranged from 1.011 to 1.025, and the HF cold-day ratio ranged from 1.043 to 1.113. Removing the most influential month preserved both directions. Later monthly lags weakened the CHD hot-night estimate. HF cold days remained elevated at a one-month lag and weakened at two months. These checks reduce concern that one coding choice or observation created the pattern, but they do not establish a daily lag sequence.

The attempted monthly-outcome/daily-exposure estimator passed its numerical convergence checks but failed its substantive calibration requirements. In the post-run F1.2 worst-cell re-summary, false-positive rates reached 0.150, confidence-interval coverage fell to 0.840, and sign recovery was poor for moderate effects. The method was not applied to produce a real daily coefficient.

4. Interpretation

4.1 What the study suggests

The results suggest that the way thermal burden is defined matters. Monthly average temperature and counts of official extremes did not produce the same patterns. In the CHD series, hot nights stood out more than mean temperature or very hot days. In the HF series, cold days stood out more than hot nights. This is scientifically plausible: nighttime heat may limit physiological recovery, while cold may increase circulatory strain. It is also compatible with Hong Kong research that has retained cold as an important cardiovascular hazard while drawing increasing attention to nighttime heat (Goggins, Chan, and Yang 2013; Guo et al. 2024).

Compatibility is not confirmation. The twelve models were specified after the outcome series was available, and no association survived correction across the panel. The CHD hot-night interval changed its null-crossing status across uncertainty methods. Correlated heat measures also redistributed the CHD coefficient when entered together. The defensible conclusion is therefore that the panel contains two hypotheses worth testing with better data, not two causal discoveries.

4.2 Why the result still matters

Null or uncertain findings can be useful when they close off weak claims. This project establishes that monthly governed data can support a transparent comparison of thermal-burden measures. It also shows where that design stops. The data cannot identify within-month timing, individual exposure, or the cause of admission. The failed daily-exposure calibration provides an empirical reason not to present a daily coefficient, rather than relying only on a general warning about aggregation.

The distinction has practical consequences. A health-protection message based on confirmed daily triggering would require different evidence from an exploratory monthly burden pattern. The present findings do not justify disease-specific warning thresholds. They do support continued attention to both overnight heat and winter cold, and they motivate a governed design that links daily weather to dated, cause-recorded admissions.

4.3 Strengths

The project has five main strengths. First, it uses an eleven-year public weather record and governed health aggregates. Second, weather definitions were source-checked rather than reconstructed from memory. Third, all twelve core comparisons are reported as one panel, reducing selective emphasis on the smallest p-values. Fourth, four uncertainty methods are displayed rather than selecting the one that gives the most attractive interval. Fifth, tables, figures and claim text are linked to machine-readable outputs and automated release checks.

4.4 Limitations

The limitations determine how the results should be read:

1. **Outcome meaning.** Admission cause was not recorded. The event is the first hospitalisation after a recorded CHD or HF diagnosis, not necessarily an admission caused by that condition.
2. **Ecological resolution.** Each observation describes Hong Kong for one month. Individual exposure and within-month timing are unknown, so individual or daily causal inference is unavailable ([Greenland and Morgenstern 1989](#)).
3. **Denominator.** The number of cohort members still eligible for a first event in each month was not available. The estimates are count ratios, not cohort incidence-rate ratios.
4. **Confounding.** Season and long-term trend were controlled, but the core panel does not claim complete adjustment for pollution, humidity, influenza, behavioural change, or changes in health care.
5. **Exposure measurement.** Headquarters weather is a territory-level proxy. It cannot represent differences between neighbourhoods, indoor temperatures, air-conditioning access, or personal activity.
6. **Serial dependence and multiplicity.** CHD residuals remained correlated over time, and no core comparison survived correction for twelve tests.
7. **Available outcomes.** Age, sex and disease subgroups were unavailable. A stroke series was not delivered and no stroke result is reported.

5. Implications and next study

The next analysis should not simply add more monthly definitions. Its highest value would come from improving the outcome data contract. A stronger design would include the dated admission, recorded reason for admission, cohort entry and follow-up, and the number of people still at risk of a first event. Daily or weekly outcome resolution would make lag analysis possible. Age, sex, diagnostic subtype and location

would allow assessment of vulnerability and exposure misclassification. Pollution, humidity and influenza could then be incorporated in a clearly staged confounding plan.

The two exploratory patterns provide specific hypotheses for that design: whether repeated hot nights precede CHD-related admissions, and whether cold days precede HF-related admissions among people with diabetes or hypertension. Those hypotheses should be declared before fitting the next outcome series. The analysis should retain both heat and cold measures, use a limited primary contrast set, and preserve the full uncertainty display.

For public-health practice, the immediate message is deliberately broader. Hong Kong’s changing climate includes more nighttime heat without the elimination of cold. Cardiovascular adaptation should therefore avoid a heat-only narrative. This report does not quantify preventable admissions, but it identifies the evidence needed before burden estimates or disease-specific recommendations can be defended.

6. Conclusion

Between 2013 and 2023, Hong Kong experienced increasing hot-night burden while cold days persisted. In governed monthly data for people with diabetes and/or hypertension, CHD first-hospitalisation counts were most elevated in relation to hot nights, while HF counts were most elevated in relation to cold days. Neither pattern survived correction across twelve comparisons, and the CHD hot-night result was sensitive to the treatment of uncertainty.

The study therefore offers hypotheses, not proof of thermal effects. Its more durable contribution is a reproducible and aggregation-aware framework: define the event precisely, compare thermal measures as a panel, show uncertainty, and decline daily or causal claims when the data cannot support them.

Acknowledgements

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Appendix A. Complete core panel

Table A1. Separate negative-binomial monthly models with a days-in-month offset and Newey-West lag-6 confidence intervals.

Outcome	Exposure contrast	Count ratio (95% CI)	BH α
CHD	Mean temperature / 1 °C	0.993 (0.976–1.010)	0.726
CHD	Mean maximum temperature / 1 °C	0.993 (0.979–1.007)	0.642
CHD	Mean minimum temperature / 1 °C	0.994 (0.977–1.012)	0.764
CHD	Hot nights / 5 days	1.022 (1.002–1.042)	0.192
CHD	Cold days / 5 days	0.995 (0.949–1.043)	0.900
CHD	Very hot days / 5 days	0.999 (0.974–1.025)	0.962
HF	Mean temperature / 1 °C	0.974 (0.947–1.002)	0.207
HF	Mean maximum temperature / 1 °C	0.981 (0.958–1.005)	0.272
HF	Mean minimum temperature / 1 °C	0.973 (0.947–1.000)	0.202
HF	Hot nights / 5 days	1.003 (0.976–1.031)	0.900
HF	Cold days / 5 days	1.073 (1.006–1.144)	0.192
HF	Very hot days / 5 days	0.995 (0.963–1.028)	0.900

Appendix B. Submission interpretation

This report is an accessible research essay for the Laidlaw Stage 3 submission. It presents current CHD/HF results and supersedes the earlier literature-and-methods essay as the submission-facing research report. The journal manuscript and technical supplement contain fuller diagnostics. The separate HKU report form requires supervisor completion and endorsement.

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